



UNIVERSITY OF
BIRMINGHAM

Superconductivity

Lecture 13 – “Josephson Junction”

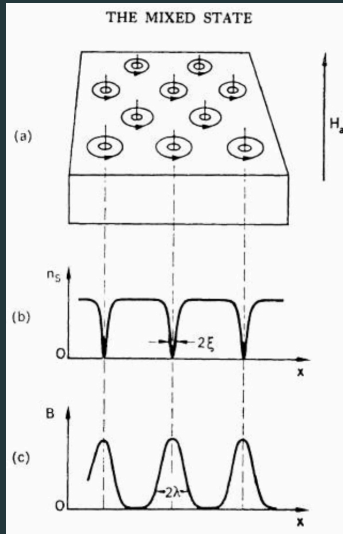
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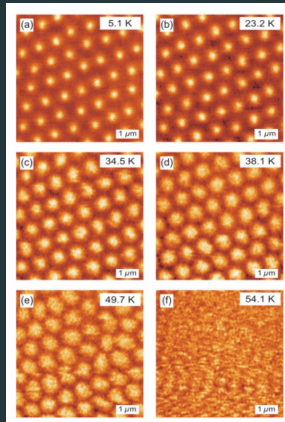
Condensed Matter Physics Group

School of Physics & Astronomy

Previously, vortices (flux lines)

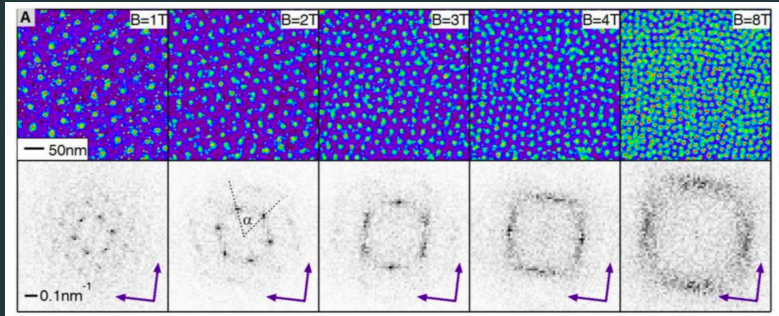


Previously, flux line lattice: T evolution



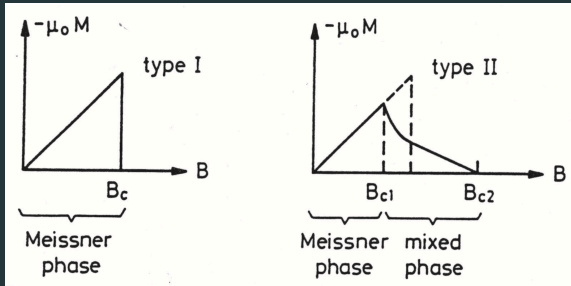
• $\lambda \simeq 100 \text{ nm}$.

Previously, flux line lattice: H evolution



• $\xi \simeq 5\text{ nm}$.

Previously, magnetisation curve



Mind the gap

Two superconductors separated

Two superconductors separated by a small distance (e.g., by insulating layers).

Their order parameters:

$$\begin{aligned}\psi_1 &= |\psi_1|e^{i\theta_1} = \sqrt{n_{s1}/2}e^{i\theta_1}, \\ \psi_2 &= |\psi_2|e^{i\theta_2} = \sqrt{n_{s2}/2}e^{i\theta_2}.\end{aligned}$$

- K = coupling between the two superconductors;
- V = potential diff,

One may write

$$\begin{aligned}i\hbar\frac{\partial}{\partial t}\psi_1 &= E_0\psi_1 + K\psi_2, \\ i\hbar\frac{\partial}{\partial t}\psi_2 &= (E_0 - 2eV)\psi_2 + K\psi_1,\end{aligned}$$

Coupled differential equations.

Constructing differential equation

$$\begin{aligned}\frac{\partial}{\partial t}\psi &= \frac{\partial}{\partial t}(|\psi|e^{i\theta}) \\ &= \left(\frac{1}{2n_s}\frac{\partial n_s}{\partial t} + i\frac{\partial\theta}{\partial t}\right)\psi.\end{aligned}$$

Then one can rewrite

$$\begin{aligned}i\hbar\frac{\partial}{\partial t}\psi_1 &= E_0\psi_1 + K\psi_2, \\ i\hbar\frac{\partial}{\partial t}\psi_2 &= (E_0 - 2eV)\psi_2 + K\psi_1,\end{aligned}$$

as

$$\begin{aligned}\left(i\hbar\left(\frac{1}{2n_{s1}}\frac{\partial n_{s1}}{\partial t} + i\frac{\partial\theta_1}{\partial t}\right) - E_0\right)\sqrt{n_{s1}}e^{i\theta_1} &= K\sqrt{n_{s2}}e^{i\theta_2}, \\ \left(i\hbar\left(\frac{1}{2n_{s2}}\frac{\partial n_{s2}}{\partial t} + i\frac{\partial\theta_2}{\partial t}\right) - E_0 + 2eV\right)\sqrt{n_{s2}}e^{i\theta_2} &= K\sqrt{n_{s1}}e^{i\theta_1}\end{aligned}$$

Josephson equations

Solve separately for real and imaginary parts.

1. From equating the imaginary parts:

$$\begin{aligned}\frac{dn_{s1}}{dt} &= -\frac{dn_{s2}}{dt} \\ &= \frac{2K}{\hbar} \sqrt{n_{s1}n_{s2}} \sin(\theta_2 - \theta_1).\end{aligned}$$

2. From equating the real parts:

$$\frac{d}{dt}(\theta_2 - \theta_1) = \frac{2eV}{\hbar}.$$

DC Josephson effect

$$\begin{aligned}\frac{dn_{s1}}{dt} &= -\frac{dn_{s2}}{dt} \\ &= \frac{2K}{\hbar} \sqrt{n_{s1}n_{s2}} \sin(\theta_2 - \theta_1).\end{aligned}$$

Current density

$$\begin{aligned}j &= -e \left(\frac{dn_{s2}}{dt} - \frac{dn_{s1}}{dt} \right) \\ &= \frac{4Ke}{\hbar} \sqrt{n_{s1}n_{s2}} \sin(\theta_2 - \theta_1) \\ &= j_0 \sin(\theta_2 - \theta_1).\end{aligned}$$

- Current flows without a voltage
- Current is driven by the phase difference.

Current-voltage characteristic

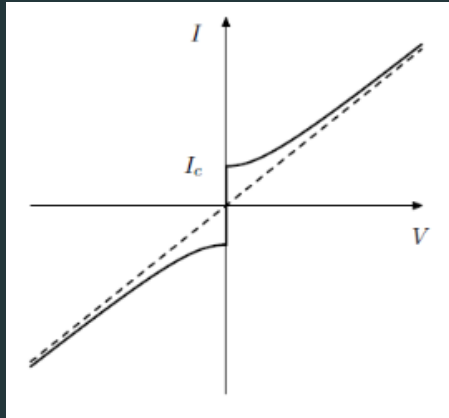


Figure 1: $I - V$ curve. I_c is the critical current for the Josephson junction (in the previous slide, it was denoted as j_0 ; sorry for mix in notation)

AC Josephson effect (1/2)

$$\frac{d}{dt}(\theta_2 - \theta_1) = \frac{2eV}{\hbar}.$$

Let's denote $\theta_2 - \theta_1 = \theta_{21}$,

$$\theta_{21}(t) = \theta_{21}(0) + \frac{2e}{\hbar}Vt.$$

An applied 'constant' voltage increases the phase difference. According to the DC Josephson effect,

$$I(t) = I_0 \sin \left(\theta_{21}(0) + \frac{2e}{\hbar}Vt \right).$$

$\Rightarrow$ AC current sets in.

AC Josephson effect (2/2)

The AC current

$$I(t) = I_0 \sin \left(\theta_{21}(0) + \frac{2e}{\hbar} V t \right).$$

oscillates with a frequency

$$f = \frac{2e}{h} V,$$

which is used as a voltage standard.

Microwave irradiation

Suppose

$$V(t) = V_0 + V_1 \cos(\omega_1 t).$$

This leads to

$$I(t) = I(0) \sin \left(\theta_{21}(0) + \omega_0 t + \frac{2eV}{\hbar\omega_1} \sin(\omega_1 t) \right),$$

which can be expanded using Bessel functions.

A DC contribution only when

$$\omega_0 = n\omega_1,$$

Shappiro steps

$$\omega_0 = n\omega_1,$$

equivalent to

$$V_0 = n \frac{\hbar \omega_1}{2e}$$

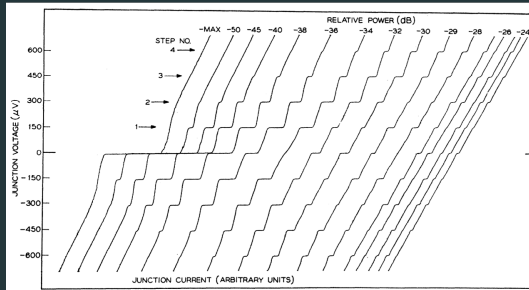
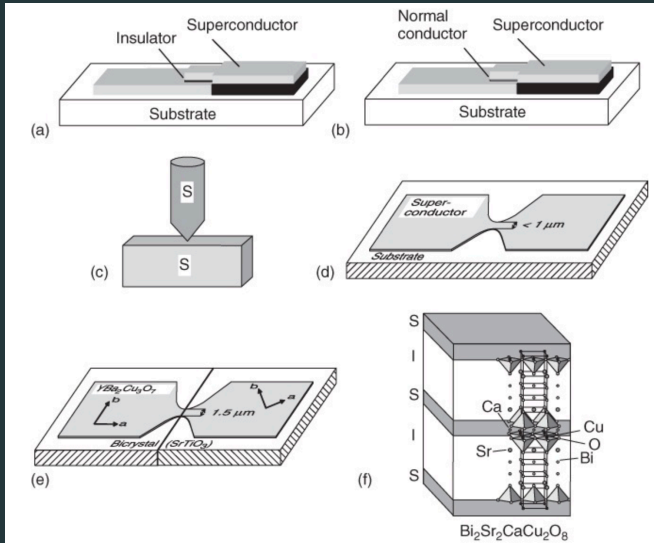
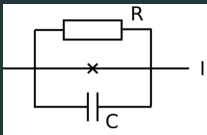


Figure 2: IV characteristic of a Nb-Nb point contact Josephson junction.

Examples of Josephson junction



Physical (realistic) Josephson junction



RCSJ (Resistively and Capacitively Shunted Junction) model.

By equating $I = I_R + I_C + I_J$ where I_J is the Josephson current, with the AC Josephson equation, one obtains,

$$\frac{\hbar C}{2e} \frac{d^2 \phi}{dt^2} + \frac{\hbar}{2eR} \frac{d\phi}{dt} + I_c \sin \phi = I. \quad (1)$$

One can show (for large C)

$$V = R(I^2 - I_c^2)^{1/2} \quad (2)$$